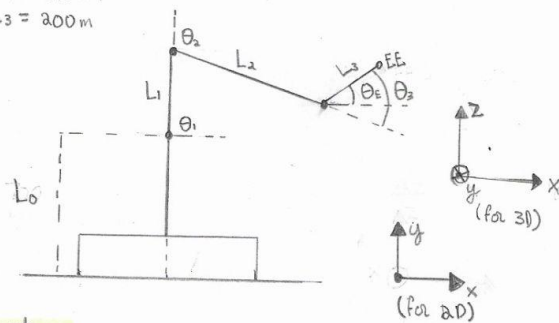


- By ignoring the first axis (waist) and the fifth axis (wrist rotation), we get a planar (2D) robot with 3 degrees of liberty. We go from (x, y, z) to (x, y) .

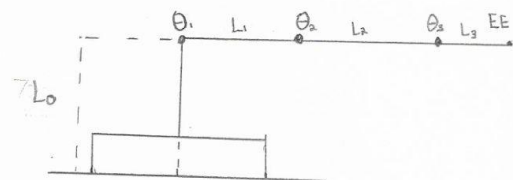
Representation of the (planar) robot

$$\begin{aligned} L_1 &= 450 \text{ mm} & L_0 &\approx 700 \text{ mm} \\ L_2 &= 670 \text{ mm} \\ L_3 &= 200 \text{ mm} \end{aligned}$$



Representation of the planar robot in the relaxed position ($\theta_1 = \theta_2 = \theta_3 = 0$)

* some configurations (like this one) are impossible due to the angle value constraints.



DH parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3
4	0	L_3	0	0

Joint angle properties

- θ_1 (shoulder), range: $[-40^\circ, 40^\circ] \rightarrow$ range: $[130^\circ, 50^\circ]$
- θ_2 (elbow), range: $[\approx 10^\circ, \approx -35^\circ] \rightarrow$ range: $[\approx 10^\circ, \approx -35^\circ]$
- θ_3 (wrist bending), range: $[-90^\circ, 90^\circ] \rightarrow$ range: $[-90^\circ, 90^\circ]$

Joint angle ranges

Method (Part #1)

- Calculating the position of the 3rd joint via θ_{EE} and (x_{EE}, y_{EE}, z_{EE}) which are provided by the user

$$\begin{aligned} x_3 &= x_{EE} - L_3 \cdot \cos(\theta_{EE}) \\ y_3 &= y_{EE} - L_3 \cdot \sin(\theta_{EE}) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{We are now left with a 2DOF robot}$$

$$\rightarrow {}^0_3T = \begin{bmatrix} C_{123} & -S_{123} & 0 & L_1 \cdot S_1 + L_2 \cdot C_{12} \\ S_{123} & C_{123} & 0 & L_1 \cdot S_1 + L_2 \cdot S_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \left. \begin{array}{l} \rightarrow x_3 = L_1 \cdot C_1 + L_2 \cdot C_{12} \\ \rightarrow y_3 = L_1 \cdot S_1 + L_2 \cdot S_{12} \\ \rightarrow \theta_{EE} = \theta_1 + \theta_2 + \theta_3 \end{array} \right\} 3 \text{ EQ, 3 Unknowns}$$

$$\rightarrow x_3^2 + y_3^2 = L_1^2 + L_2^2 + 2L_1L_2C_2 \rightarrow C_2 = \frac{x_3^2 + y_3^2 - L_1^2 - L_2^2}{2L_1L_2} \rightarrow S_2 = \pm \sqrt{1 - C_2^2}$$

$\rightarrow$ 2 possibilities for S_2

$$\begin{aligned} \theta_2 &= \text{atan2}(S_2, C_2) \\ \text{or} \\ \theta_2 &= \text{atan2}(-S_2, C_2) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Elbow angle}$$

Knowing θ_2 , we can now calculate θ_1 .

$$\rightarrow X_3 = L_1 \cdot C_1 + L_2 \cdot C_{12} \rightarrow X_3 = L_1 \cdot C_1 + L_2 (C_1 C_2 - S_1 S_2)$$

$$\rightarrow Y_3 = L_1 \cdot S_1 + L_2 \cdot S_{12} \rightarrow Y_3 = L_1 \cdot S_1 + L_2 (C_1 S_2 + S_1 C_2)$$

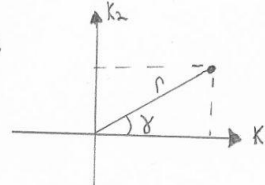
From now, we will work with X_3 , but we could also have selected Y_3

$$\rightarrow X_3 = L_1 \cdot C_1 + L_2 C_2 C_1 - L_2 S_2 S_1$$

$$X_3 = K_1 \cdot C_1 - K_2 \cdot S_1 \quad (1) \rightarrow \text{Where: } K_1 = L_1 + L_2 C_2$$

$$K_2 = L_2 \cdot S_2$$

$$\rightarrow Y_3 = K_1 \cdot S_1 + K_2 \cdot C_1 \quad (2)$$



$$r = \sqrt{K_1^2 + K_2^2}$$

$$\gamma = \text{atan2}(K_2, K_1)$$

$$\downarrow$$

$$K_1 = r \cdot \cos(\gamma) \quad (3)$$

$$K_2 = r \cdot \sin(\gamma)$$

$$\rightarrow \text{Using (1), (2), (3): } \rightarrow \frac{X_3}{r} = \cos(\gamma) \cos(\theta_1) - \sin(\gamma) \sin(\theta_1) = \cos(\gamma + \theta_1)$$

$$\rightarrow \frac{Y_3}{r} = \cos(\gamma) \sin(\theta_1) + \sin(\gamma) \cos(\theta_1) = \sin(\gamma + \theta_1)$$

$$\rightarrow \gamma + \theta_1 = \text{atan2}\left(\frac{Y_3}{r}, \frac{X_3}{r}\right)$$

$$\rightarrow \theta_1 = \text{atan2}(Y_3, X_3) - \text{atan2}(K_2, K_1) \quad \text{shoulder angle}$$

$$\rightarrow \theta_3 = \theta_{EE} - \theta_1 - \theta_2 \quad \text{wrist bending angle}$$

Recap and other angles (wrist and wrist)

→ The program input to calculate the inverse kinematics: θ_{EE} and (X_{EE}, Y_{EE}, Z_{EE})

→ Computing the (shoulder, elbow and wrist bending angles):

$$\rightarrow X_{EE-2D} = \sqrt{X_{EE}^2 + Y_{EE}^2}$$

$$\rightarrow Y_{EE-2D} = Z_{EE} - L_0$$

$$\rightarrow X_{3-2D} = X_{EE-2D} - L_3 \cdot \cos(\theta_{EE}) =$$

$$\rightarrow Y_{3-2D} = Y_{EE-2D} - L_3 \cdot \sin(\theta_{EE}) =$$

$$\rightarrow C_2 = \frac{X_{3-2D}^2 + Y_{3-2D}^2 - L_1^2 - L_2^2}{2 L_1 L_2}$$

$$\rightarrow S_2 = \pm \sqrt{1 - C_2^2}$$

$$\left. \begin{array}{l} C_2 \\ S_2 \end{array} \right\} \theta_2 = \text{atan2}(S_2, C_2) \rightarrow \text{elbow angle}$$

$$\rightarrow K_1 = L_1 + L_2 C_2$$

$$\rightarrow K_2 = L_2 \cdot S_2$$

$$\left. \begin{array}{l} K_1 \\ K_2 \end{array} \right\} \theta_1 = \text{atan2}(Y_{3-2D}, X_{3-2D}) - \text{atan2}(K_2, K_1) \rightarrow \text{shoulder angle}$$

$$\rightarrow \theta_3 = \theta_{EE} - \theta_1 - \theta_2 \rightarrow \text{wrist bending angle}$$

$$\rightarrow \theta_{WA} = \text{atan}(Y_{EE}, X_{EE}) \rightarrow \text{wrist angle}$$

→ θ_{WR} : The wrist rotation angle has no effect on the end effector - can be set at random